

HACKATHON BUILD BRIEF · V1.0

QUORUM.

Every AI advisor agrees with you. That's the bug. Quorum makes them fight — with real workflows, not just prompts.



FORMAT

CORE IDEA

ADVISORS

SURFACE

Hackathon MVP **Multi-Perspective Decision Engine** **5 Workflow Agents** **Spatial Canvas + Cross-Chat**

1 What We're Building (and Why It Demos Well)

Condensed from the full PRD — same product vision, scoped for a hackathon clock

Quorum is a self-hostable decision engine. A user asks one question — *"Should we launch a freemium tier?"* — and five AI Advisors (CMO, CFO, CTO, COO, Contrarian) each run a **distinct multi-step workflow** — not the same prompt with a different persona — and land as color-coded cards on a live spatial canvas. The user drags cards together to spark a cross-chat, and the panel actually disagrees with itself.

For a hackathon, this is a strong pick *because* the "wow" is visual and immediate: judges see five distinct research trails converge in real time, watch a contrarian agent attack the consensus, and drag two cards together to watch them argue. That is demoable in 90 seconds without narration.

The Three Pillars (unchanged from the PRD — this is the whole pitch)

01

Workflow-Driven Agents

Each Advisor runs retrieve → research → analyze → compile. Real tool calls, not persona theater.

02

Canvas-Native Interaction

Cards live on a free-drag 2D surface. Connect 2-3 cards to spawn a focused cross-chat.

03

Persistent Memory

Decision Records persist; future debates cite past ones. Canvases save and resume.

What Existing "AI Advisor" Demos Get Called Out For — and How We Avoid It

✗ **Persona theater**

Five system prompts, one LLM call each, same generic answer reworded.

✓ **Quorum**

Every Advisor has a logged, different tool-using pipeline — judges can open the trace.

✗ **Sycophantic convergence**

All agents politely agree; nothing to actually watch happen on stage.

✓ **Quorum**

Contrarian is contractually forbidden to agree — guaranteed on-stage conflict.

✗ **Linear chat jail**

A scrolling thread — nothing spatial to show a judge from across the table.

✓ **Quorum**

A literal canvas: drag, connect, watch a popup conversation spawn live.

✗ **No memory**

Nothing persists between runs — trivially replicated by any wrapper.

✓ **Quorum**

Decision Records save and get cited in the next debate — show it live, twice.

2 The Advisor Panel — All 5 Workflows, Kept Intact

Full pipelines preserved from the PRD; step count is what makes this defensible, not decorative

CMO Chief Marketing Officer

Rubric: market positioning, demand generation, differentiation. Must attack claims without customer evidence.

1. Understand initiative →
2. Retrieve marketing KB →
3. Web search: competitors →
4. Channel & positioning analysis →
5. Campaign strategy draft →
6. Compile card

LLM calls: 3**Tools:** web_search**KB scope:** marketing, general

CFO Chief Financial Officer

Rubric: financial viability, margin protection, cash timing. Must attack unpriced costs and optimistic revenue.

1. Extract financial dimensions →
2. Retrieve P&L / financial data →
3. Quantitative model (calculator) →
4. Risk & scenario stress-test →
5. Compile card

LLM calls: 3**Tools:** table_query / calculator**KB scope:** finance, general

CTO Chief Technology Officer

Rubric: technical feasibility, scalability, build-vs-buy. Must attack underestimated complexity and tech debt.

1. Extract technical dimensions →
2. Retrieve tech/architecture context →
3. Web search: landscape research →
4. Feasibility & architecture analysis →
5. Compile card

LLM calls: 3**Tools:** web_search**KB scope:** product, ops, general

COO Chief Operating Officer

Rubric: execution feasibility, resource capacity, timeline realism. Must attack plans that assume too much bandwidth.

1. Extract operational dimensions →
2. Retrieve ops/team context →
3. Capacity & dependency mapping →
4. Execution plan & risk analysis →
5. Compile card

LLM calls: 3**Tools:** none (internal-data-heavy)**KB scope:** ops, general

CONTRARIAN Devil's Advocate · runs last, on purpose

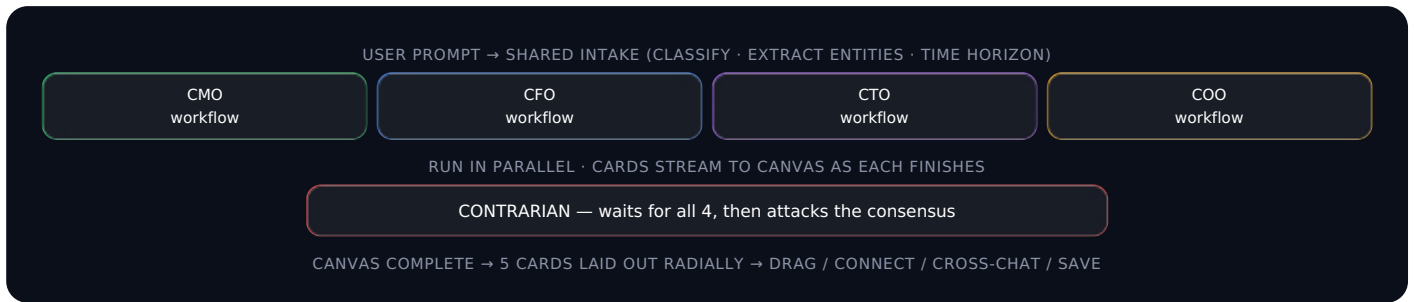
Rubric: contractually forbidden to agree. Must surface the riskiest shared assumption and a historical failure analogy.

1. Wait for all 4 peer cards →
2. Identify consensus & assumptions →
3. Web search: failure patterns →
4. Build the steelmanned counter-case →
5. Compile card

LLM calls: 3**Tools:** web_search**KB scope:** ALL (full read access)

3 Orchestration & the Signature Canvas Interaction

This is what's on screen when judges walk up — keep it exactly this visual



Canvas Interaction — The Non-Negotiable Demo Beat

- 1. Cards stream in live.** Each card shows a progress line ("Researching competitors... → Analyzing... → Compiling...") before the verdict types in. This alone sells "real workflow, not a spinner."
- 2. Drag two cards together.** A connection line renders and a floating Cross-Chat popup opens automatically — canvas stays visible behind it.
- 3. Ask a pointed question.** "CMO says 22% cannibalization — CFO, does the model support that number?" Advisors respond in turns, citing evidence, max 6 turns.
- 4. Generate a merged card.** The reconciled (or explicitly unresolved) conclusion appears between the two parents with a blended color strip.

Judge line to say out loud: "Every advisor cites a real source — a KB chunk, a calculation, or a web result — never a vibe. The Contrarian is the only agent not allowed to agree with the room."

Card Anatomy (unchanged from PRD — keep every element, it's what makes cards feel "real")

- ✓ Color strip + Advisor name + collapse/resize controls
- ✓ Bold one-line verdict (with a number in it for CFO)
- ✓ Markdown body: analysis, recommended plan, risks
- ✓ Clickable citation chips → source drawer (KB chunk / web link / Decision Record)
- ✓ Confidence label: Grounded / Partial / Speculation — mandatory, never omitted
- ✓ Assumptions list
- ✓ Action bar: Connect · Ask this Advisor · Pin

4 Hackathon Build Plan — Scoped to the Clock

All PRD features are kept in scope; this reorders them into what must exist to demo vs. what impresses if time allows

MUST DEMO Core Loop — non-negotiable

- Seeded fixture Knowledge Base (fake company: P&L CSV, marketing doc, ops doc, roadmap doc)
- All 5 Advisor workflows run for real — every step logged, no collapsing into one mega-prompt
- Cards stream onto a canvas with live progress indicators
- Drag-to-connect → Cross-Chat popup → at least one full back-and-forth → merged card
- Citations are real and clickable (KB chunk / web result), confidence label always shown

SHOULD HAVE Strengthens the pitch

- Canvas save/resume + a visible Canvas Gallery with 2+ pre-made saved canvases
- Decision Record generated at the end of a debate, saved to KB
- A second debate in the same demo that visibly cites the first Decision Record
- Card states: compact / minimized, for canvas decluttering
- Summary Report generation (select cards → one synthesized report card)

STRETCH If the clock allows

- Custom Advisor workflow editor (add/reorder steps for a 6th, user-made Advisor)
- Card grouping/bundling and pin-to-top
- Feed-view (linear) fallback for mobile/accessibility
- Cost meter + per-debate budget cap shown live in the header
- An Advisor Pack (YAML) import, e.g. a "Legal Advisor" pack, demoed live

Suggested Timeline (24–36 hr hackathon)

Pre-event

Scaffold & fixtures

Monorepo, shared Zod schemas, LLM provider wrapper, and the fixture KB (P&L CSV + 3 docs) — do this before the clock starts if allowed.

Hr 0–4

One workflow, end to end

Get CFO's full 5-step pipeline (the hardest one — real calculator tool) working and rendering as a single card. Prove the pattern before replicating it.

Hr 4–10

All 5 Advisors + linear grid

Clone the pattern for CMO, CTO, COO, Contrarian. Ship a plain responsive grid UI first — no canvas yet. This is your fallback demo if canvas breaks.

Hr 10–18

Canvas + connect + cross-chat

Swap the grid for a react-flow canvas. Implement drag-to-connect and the floating cross-chat popup. This block is the signature UX — protect the time for it.

Hr 18–24

Decision Records + polish

Wire Decision Record save/recall so a second debate visibly cites the first. Clean up empty states, confidence labels, and the empty-KB watermark.

Hr 24+

Stretch features + rehearsal

Only touch stretch items once the must-demo loop is rock solid. Rehearse the 90-second demo path at least three times.

5 Tech Stack & Data Model — Fast-to-Build Choices

Same stack as the full PRD; chosen because it needs zero infra setup on the day

Layer	Choice	Why it's hackathon-friendly
Frontend	React 19 + TypeScript + Vite · @xyflow/react for canvas · Tailwind · Zustand · SSE streaming	@xyflow/react gives drag/connect/pan/zoom for free — do not hand-roll canvas physics
Backend	Node + TypeScript (Fastify or Hono)	Minimal boilerplate, SSE support out of the box
Database	SQLite + Drizzle ORM (+ FTS5 + sqlite-vec)	Zero-dependency, no hosted DB to provision
Engine	Hand-rolled async workflow orchestrator (no heavy agent framework)	Determinism & visible tracing matter more than framework features for a demo
Tools	web_search (Serper/Tavily), table_query (sandboxed calc over CSV), dr_lookup	Three small, testable tool functions — build once, reuse across Advisors
Parsing	pdf-parse / mammoth / papaparse	Covers the fixture KB formats without extra setup

Minimum Data Model (trim from PRD §7 — keep only what the demo touches)

Table	Key fields
advisors	name, color, workflow_steps_json, rubric, data_scope[], tools[]
debates	question, intake_json, status, cost_usd
cards	advisor_id, verdict, body_md, claims_json, assumptions_json, confidence, position_x/y
connections / cross_chats	card_ids[], transcript_json, merged_card_id
decision_records	question, chosen_option, rationale_md, assumptions_json, review_date
workflow_traces	steps_json[{step_name, input, output, tokens, cost, latency}] — powers the "prove it's real" moment

Non-Negotiables Carried Over From the Full PRD

- ✓ Never collapse an Advisor's multi-step pipeline into one long prompt — judges can ask to see the trace
- ✓ Contrarian always runs last and always reads the other four cards
- ✓ Cross-chat is a floating popup, never a full-page takeover — canvas stays visible
- ✓ Every claim carries a citation or is explicitly labeled speculation
- ✓ Confidence label and empty-KB watermark are mandatory, not cosmetic

Demo safety net: Ship the Phase-1 linear grid view even after canvas works — if the canvas breaks live, fall back to the grid without losing the "5 real workflows" story.